NATIONAL UNIVERSITY
BANGLADESH



First Year Syllabus

Department of Physics

Four Year B.Sc. (Honours) Program

Effective from the Session: 2024-25

Mission

To promote "Progress for Everyone" by fostering transformative education, conducting pioneering research, and encouraging meaningful collaboration with individuals, communities, and partners both in Bangladesh and around the world.

Vision

We expect that the hundredth anniversary of the National University, our commitment to "Progress for Everyone" will be evident in all areas of our operations and stakeholder engagements.

We leverage our core strength in education, research, and community engagement to address pressing global issues and foster sustainable development.

By 2042, National University aims to strengthen its positions as Bangladesh's top academic institution and rank among the world's leading universities. Our focus will be on expanding expertise building strategic partnerships, and increasing our global influence. Our services will be measured by the achievements of the individuals and communities we serve.

Mission of the Program:

The academic program is focused on the creation, translation, and dissemination of knowledge on the subject matter. The strategic goals of the department are to:

1. Support the aims and objectives of the University within the capacity of our Departmental program. Curriculum for BSc (Hons) in Physics,
2. Train and produce high-quality graduates to meet up national and international requirements in scientific sectors of the job market.
3. Enhance the teaching-learning and research capacities of the Department by retaining and recruiting outstanding faculty and staff.

Program Educational Objectives (PEO):

PEO-1: To prepare physics graduates to exhibit quality of excellence, critical thinking, creativity, inventiveness, and self-motivation for life-long learning to handle all kind of diverse situations in interdisciplinary and multidisciplinary environment.

PEO-2: To produce graduates who are globally acceptable professionals for government, corporate and research organizations along with skills for entrepreneurial pursuits in multidisciplinary areas.

PEO-3: To groom graduates who can demonstrate technical competence in the field of project and technology management.

PEO-4: To produce graduates who can ethically lead and work as a part of team towards the fulfillment of both individual and organizational goals.

Program Learning Outcomes (PLO):

On successful completion of these programs a student should able to learn the following outcomes:

Fundamental Skill:

PLO-1: Knowledge: Apply fundamental knowledge of physics along with specializations to solve scientific problems of material science, energy, mechanics, optics, communication, space, biomedical physics etc.

PLO-2: Problem analysis: Solve problems in Physics using appropriate mathematical tools.

PLO-3: Experimentation: An ability to execute and analyze the results of an experimental investigation or theoretical modeling and to draw valid conclusions with an estimate of the uncertainty in the result. An ability to compare experimental results with the predictions of relevant theories.

PLO-4: Conduct investigations of problems: Use research based methods including design of experiments, analysis and interpretation of data leading to logical conclusions.

PLO-5: Modern tool usage: Create, select, and apply appropriate techniques, resources, software and modern scientific and ICT tools to problem solving in different sectors of Physics.

Social skill:

PLO-6: Communication: An ability to communicate scientific information verbally and in the form of clear and accurate scientific reports.

PLO-7: Environment and sustainability: Understand the impact of the professional scientific solutions in the societal and environmental contexts, and demonstrate the knowledge of, and the need for sustainable developments.

PLO-8: Ethics: Apply ethical principles and commit to professional ethics, responsibilities and norms of scientific practices.

Personal Skill:

PLO-9: Individual and teamwork: Function effectively as an individual and as a member or leader in diverse teams, and in multidisciplinary settings.

PLO-10: Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Thinking Skill:

PLO-11: Project management: Demonstrate knowledge and understanding of principles of physics and apply those to one's own work as a member and leader in a team, to manage projects and in multidisciplinary environments.

PLO-12: Technology management: Competence in using programming languages and in troubleshooting and solving basic problems in electronic equipment and circuits.

Mapping PEO with PLO:

Domain	PLO	PEO-1	PEO-2	PEO-3	PEO-4
Cognitive	**PLO-1**	√	√		
	PLO-2	√	√		
	PLO-3			√	
	PLO-4		√	√	
	PLO-5		√	√	
	PLO-6		√	√	
Affective	**PLO-7**				√
	PLO-8				√
	PLO-9				√
	PLO-10				√
Psychomotor	**PLO-11**			√	
	PLO-12			√	

Courses and Marks Distribution

First Year

Course Code	Course Title	Marks	Credits
	Core Courses		
212701	Mechanics	100	4
212703	Properties of Matter, Waves & Oscillations	100	4
212705	Heat, Thermodynamics and Radiation	100	4
212706	Practical-I (Properties of Matter, Waves & Oscillations, Heat, Thermodynamics and Radiation)	100	4
	Any two sets from following groups		
	Group-A		
213709	Foundations of Mathematics	100	4
213711	Calculus-1	50	2
	Group-B		
212807	Chemistry-I	100	4
212808	Chemistry-I Practical	50	2
	Group-C		
213607	Fundamentals of Statistics	100	4
213610	Lab-1: Fundamentals of Statistics	50	2
	Compulsory Courses		
219901	History of Bangladesh: Language, Culture and Identity	100	4
219903	Information and Communication Technology	75	3
219904	Lab: Information and Communication Technology	25	1
	Total=	**900**	**36**

Matrix of Courses with Program Learning Outcomes (Curriculum map):

	PLO-1	PLO-2	PLO-3	PLO-4	PLO-5	PLO-6	PLO-7	PLO-8	PLO-9	PLO10	PLO11	PLO12
212701	√	√										
212703	√	√										
212705	√	√										
212706	√	√	√	√	√				√			
213709	√	√										
213711	√	√										
212807	√	√										
212808	√	√		√	√				√			
212607	√	√									√	
212610	√	√		√	√							
219901	√	√						√				
219903	√				√							
219904					√				√			

Course Code	212701	Marks: 100	Credits: 4	Class Hours: 56
Course Title:	Mechanics			

Course Objectives:

This course reviews the concepts of mechanics learnt at school from a more advanced perspective and goes on to build new concepts. It begins with Newton's Laws of Motion and ends with the Forces and Special Theory of Relativity. Students will also appreciate to learn the Collisions in CM Frame, Gravitation, Rotational Motion and Oscillations. The students will be able to apply the concepts learnt to several real world problems.

Course Learning Outcomes (CLO):

After going through the course, the student should be able to learn the following outcomes

CLO1 Understand concept of quantities and units

CLO2 Grasp the knowledge of one- two- and three- dimensional motion and their applications in real life problems

CLO3 (i) Understand laws of motion and their applications. He / she will learn the concept of momentum, angular momentum to apply them to basic problems

(ii) Describe how fictitious forces arise in a non-inertial frame

CLO4 Understand work, energy, power efficiency, conversion of energy and Law of conservation of energy

CLO5 **(i)** Understand the phenomena of collisions and idea about center of mass and laboratory frames and their correlation. Write the expression for center of mass of a rigid body and about a given axis of symmetry for different uniform mass distributions and

(ii) Explain the concept of law of conservation of momentum and Elastic and non elastic collision

CLO6 Understand rotational kinematics

CLO7 Understand the analogy between translational and rotational dynamics, and application of both motions simultaneously in analyzing rolling with slipping

Mapping of CLOs with PLOs

	PLO-1	PLO-2	PLO-3	PLO-4	PLO-5	PLO-6	PLO-7	PLO-8	PLO-9	PLO10	PLO11	PLO12
CLO1	√	√										
CLO2	√	√										
CLO3	√	√										
CLO4	√	√										
CLO5	√	√										
CLO6	√	√										
CLO7	√	√										

Course Contents:

	Topic	Teaching Learning Strategy	CLOs
1	**Vector Algebra:** Vector and scalar quantities; Vectors and their components, Vector addition and subtraction, Scalar and vector triple products, scalar and vector fields, Vector differentiation and integration, Gradient, Divergence and Curl and their physical significance, Gauss's divergence theorem, Green's theorem and Stoke's theorem, Polar, Spherical and Cylindrical co-ordinates.	Lecture, Assignment	CLO1
2	**Concept of Measurement:** Different Measurement units, International system of units, Origin of Length mass and time, Conversion of units from one system to another.	Lecture, Assignment	CLO1
3	**Particles Motion in one dimension:** Concept of motion and frame of reference, Position and displacement, Average velocity and average speed, Instantaneous velocity and speed, Acceleration, Constant acceleration, Equations for motion with constant acceleration, Free-fall acceleration, Equation for free-fall acceleration, Particles of physics and basic structure of atoms and nuclear.	Lecture, Group Discussion, Assignment	CLO2
4	**Particles Motion in Two and Three Dimensions:** Position and displacement using vectors, Velocity and average velocity, Acceleration and average acceleration, Equation of motion using vector, Projectile motion,	Lecture, Group Discussion, Workshop and Assignment	CLO2

	Uniform circular motion.		
5	**Force and Motion:** Newton's laws of motion and their applications, Concept of mass, Force and weight, Frictional forces and Properties of friction, Drag force and terminal speed, Forces of nature.	Lecture, Group Discussion, and Assignment	CLO3
6	**Work, Energy and Power:** Kinetic and Potential energy, Work done by constant and variable forces, Work-energy theorem, Hooke's law, Work done by a spring force, Work done by weight, Power, Gravitational potential energy, Conservation of energy.	Lecture, Group Discussion, and Assignment	CLO4
7	**System of Particles:** Center of mass of systems of particles, Center of mass of rigid bodies, Linear momentum of a particle, Linear momentum of a system of particles, Conservation of linear momentum for a system of particles.	Lecture, Assignment	CLO5
8	**Collisions of Bodies:** Collisions and its classification, Impulse and linear momentum, Elastic and inelastic collision in one dimension, Motions of the center of mass of colliding bodies.	Lecture, Group Discussion, and Assignment	CLO5
9	**Rotational Kinematics:** Translational and Rotational motion, Angular Position, Angular displacement, Angular Velocity and angular acceleration, Rotation with constant angular acceleration, Relation between linear and angular kinematics of a particles in circular motion.	Lecture, and Assignment	CLO6
10	**Rotational Dynamics:** Torque and angular momentum and their relation, Kinetic energy of rotation and rotational inertia (moment of inertia), Combined Translational and rotational motion of a rigid body, Parallel and perpendicular axes theorems of moment of inertia, calculation of moment of inertia for solids of different shapes, conservation of angular momentum. Relation between angular momentum and torque.	Lecture, and Assignment	CLO7

References:

Halliday, D., Resnick, R. and Walker, J. (2001). *Fundamentals of Physics.* 6th Edition John Willey and Sons, Inc.

Khan A. H., Ishaq. M. (2017). B.Sc. Hons. *Physics 1st Part (Bengali Version).* Ideal Books.

Sears and Zemansky's. (2015). *University Physics*: Vol-1 Mechanics, 12th Edition, Pearson.

Spiegel, M.R. (1959). Vector *Analysis*. McGraw-Hill.

Course Code	212703	**Marks: 100**	**Credits: 4**	**Class Hours: 56**
Course Title:	**Properties of Matter, Waves & Oscillations**			

Course Objectives:

This course reviews the concepts of mechanics learnt at school from a more advanced perspective and goes on to build new concepts. It includes Newton's Laws of gravitation, Kepler's laws of motion, Properties of matter and waves. Students will also appreciate the Collisions in CM Frame, Gravitation, Rotational Motion and Oscillations. The students will be able to apply the concepts learnt to several real world problems.

Course Learning Outcomes:

After going through the course, the student should be able to

CLO1: Understand Kepler's laws of planetary motion and their applications.

CLO2: Demonstrate deformation of solids and to apply them to basic problems.

CLO3: Apply molecular theory to explain surface tension, cohesive and adhesive forces

CLO4: Understand fluid dynamics, Bernoulli's equation and its application.

CLO5: Understand the basic properties of waves and sound waves.

CLO6: Describe and analyze SHM and DHM

CLO7: Understand and apply vibrational knowledge in longitudinal and transverse mode of vibration in rods and air columns

Mapping of CLOs with PLOs

	PLO-1	PLO-2	PLO-3	PLO-4	PLO-5	PLO-6	PLO-7	PLO-8	PLO-9	PLO10	PLO11	PLO12
CLO1	√	√										
CLO2	√	√										
CLO3	√	√										

CLO4	√	√										
CLO5	√	√										
CLO6	√	√										
CLO7	√	√										

Course Contents:

	Topic	Teaching Learning Strategy	CLOs
1	**Gravitation:** Kepler's Laws, Law of universal gravitation, G and its determination, Inertial and gravitational mass, Acceleration due to gravity and its variation, Measurement of acceleration due to gravity by compound pendulum and Katter's pendulum, Gravitational potential and field in simple cases, Gravitational potential energy.	Lecture, Group Discussion, and Assignment	**CLO1**
2	**Elasticity:** Hooke's Law, Elastic constants of isotropic solids, Poisson's ratio and their interrelations, Internal elastic potential energy, Experimental determination of elastic constants, Torsion of a cylinder, Bending of beams, Cantilever, Variation of elasticity with temperature.	Lecture, Group Discussion, Workshop and Assignment	**CLO2**
3	**Surface Tension:** Surface tension and surface energy, Adhesive and cohesive forces, Molecular theory of surface tension, Pressure on a curved membrane of uniform tension, Soap bubble, Capillarity, Angle of contact and its measurement, Determination of surface tension of water and mercury drop, Variation of surface tension with temperature.	Lecture, Group Discussion, and Assignment	**CLO3**
4	**Fluid Dynamics:** General concepts of fluid flow, Streamlines, Equation of continuity, Bernoulli's equation, Application of Bernoulli's equation and equation of continuity. Coefficient of viscosity, Critical velocity and Reynold's number, Poiseuille's formula and its correction, Measurement of viscosity, Variation	Lecture, Group Discussion, and Assignment	**CLO4**

	of viscosity with temperature.		
5	**Waves:** Waves and Particles, Types of waves, Transverse and Longitudinal waves, Wavelength and frequency, The Speed of a traveling Wave, Wave speed on a stretched string, Energy and power of a traveling string wave, The principle of superposition for waves, Interference of waves, Complex waves, Standing waves and Resonance.	Lecture, Group Discussion, and Assignment	**CLO5**
6	**Sound Waves:** The Speed of Sound, Propagation and speed of longitudinal waves, Traveling longitudinal waves, Standing longitudinal waves, Beats, Doppler effect.	Lecture, Group Discussion, and Assignment	**CLO5**
7	**Oscillations:** Simple harmonic motion (SHM), Energy consideration in SHM, Applications of SHM, Relation between SHM and uniform circular motion, Combinations of two SHM's, Lissajous' figures, Two-body oscillations, Damped harmonic motion, Forced oscillations and resonance, Power and intensity of wave motion.	Lecture, Group Discussion, and Assignment	**CLO6**
8	**Vibrations:** Vibrations of string, Membranes, bars, plates and air-column, Sonometer, Melde's experiment, Rectangular and circular membranes, Transverse and longitudinal vibration of rod, Air-columns in cylindrical pipes, Organ pipes, Chladni's figure.	Lecture, Group Discussion, and Assignment	**CLO7**

References:

Halliday, D., Resnick, R., & Walker, J. (2001). *Fundamentals of Physics.* 6th Edition John Willey and Sons, Inc.

Khan, A.H., & Ishaq, M. (1017). *Physics* 2nd Part (Bengali Version). Ideal Books.

Sears, F.W., Zemansky, M.W,. and Young, H.D. (2007). *University Physics.* 12th Edition, Pearson Addison Wesley.

Course Code	212705	Marks: 100	Credits: 4	Class Hours: 56
Course Title:	Heat, Thermodynamics & Radiation			

Course Objectives:

This course deals with the relationship between the macroscopic properties of physical systems in equilibrium. It reviews the concepts of thermodynamics learnt at school from a more advanced perspective and develops them further. The primary goal is to understand the fundamental laws of thermodynamics and their applications to various systems and processes. In addition, it will also give exposure to students about the Kinetic theory of gases, transport phenomena involved in ideal gases, phase transitions and behavior of real gases.

Course Learning Outcomes:

After completing the course student will be able to

CLO1: **Understand b**asic concept of heat, temperature, and thermal equilibrium
(ii) Construct different types of thermometer

CLO2: Understand kinetic theory of gas

CLO3: (i)Comprehend the basic concepts of thermodynamics, the first and the second law of thermodynamics, the concept of entropy and the associated theorems, the thermodynamic potentials and their physical interpretations.

(ii) Learn about Maxwell's thermodynamic relations.
(iii) Understand the basic aspects of kinetic theory of gases, Maxwell-Boltzman distribution law, equitation of energies, mean free path of molecular collisions, viscosity, thermal conductivity, diffusion and Brownian motion.

CLO4: Explain the mechanism of heat engine and Carnot's heat cycle.

CLO5: Understand real gas equations, Van der Waal equation of state, the Joule-Thompson effect.

CLO6: Describe the concept of black body radiation, Emissive and absorptive powers, Stefan- Boltzmann's Law, Wien's displacement law, Rayleigh-Jean's law, Planck's quantum hypothesis and applications of radiations laws.

Mapping of CLOs with PLOs

	PLO-1	PLO-2	PLO-3	PLO-4	PLO-5	PLO-6	PLO-7	PLO-8	PLO-9	PLO10	PLO11	PLO12
CLO1	√	√										
CLO2	√	√										
CLO3	√	√										
CLO4	√	√										
CLO5	√	√										
CLO6	√	√										

Course Contents:

	Topics	Teaching Learning Strategy	CLOs
1	**Heat and Temperature**: Concept of temperature, Thermal equilibrium, Measurement of low and high temperatures, The Clausius & Fahrenheit scales, Thermal expansion, Gas Thermometers, Platinum resistance thermometer, Thermocouple	Lecture, Group Discussion, and Assignment	**CLO1**
2	**Kinetic Theory of Gases:** Equation of state of an ideal gas, Equipartition of energy, Translational kinetic energy, Mean free path, Maxwell's theory of distribution of velocities, Brownian motion, Degrees of freedom & Molar Specific Heats, Van der Waals' Equation of state, Transport phenomena.	Lecture, Group Discussion, and Assignment	**CLO2**
3	**Transmission of Heat:** Conduction, Convection, Radiation, Conduction of heat in solids, Measurement of thermal conductivity of a bad conductor, Heat conduction through composite walls.	Lecture, Group Discussion, and Assignment	**CLO3**
4	**First Law of Thermodynamics:** Internal Energy, Heat and work, Isothermal and adiabatic processes, Work done by expanding gases, Statement of first law of thermodynamics and applications.	Lecture, Group Discussion, and Assignment	**CLO3**

5	**Second Law of Thermodynamics and Entropy:** The Thermodynamic temperature scale, Concept of entropy, Calculation of 6entropy change in reversible and irreversible processes, Entropy and second law of thermodynamics, Entropy and disorder. The Carnot engine, Efficiency of heat engines, Carnot's theorem, Refrigerator and air-conditioner, Clausius theorem, Clausius-Clapeyron equation.	Lecture, Group Discussion, Assignment	**CLO4**
6	**Third Law of Thermodynamics:** Nernst heat theorem, Phase rule and its uses, Third law of thermodynamics.	Lecture, Group Discussion, and Assignment	**CLO3**
7	**Thermodynamic Functions:** Thermodynamic potentials at constant volume and pressure, Maxwell's thermodynamic relations, Specific heat equations, Joule-Thomson effect and its applications.	Lecture, Group Discussion, and Assignment	**CLO5**
8	**Radiation Laws:** Concept of black body and black body radiation, Emissive and absorptive powers, Kirchhoff's law, Stefan-Boltzmann's Law, Wien's displacement law, Rayleigh-Jean's law, Planck's quantum hypothesis, Planck's law, Applications of radiations laws.	Lecture, Group Discussion, and Assignment	**CLO6**

References:

Halliday, D., Resnick, R., & Walker, J. (2001). Fundamentals of Physics. 6th Edition John Willey and Sons, Inc

Hossain, T. (1975) *Text Book of Heat.* Variety Books.

Khan, A.H., and Ishaq, M.. (2017). *B.Sc. Hons Physics 3rd Part*. (Bengali Version) Ideal Books.

Saha, M.N., and Srivastava, B.N. A (1958) *Treatise on Heat.* The Indian Press Private Ltd.

Sears, F.W. (1950). An Introduction to Thermodynamics. Addison-Wesley Press

Zemansky, M.W. (1997) *Heat and Thermodynamics.* 7th Edition The McGraw-Hill Co. Inc.

Course Code	212706	Marks: 100	Credits: 4	Class Hours: 56
Course Title:	**Practical-I** (Properties of Matter, Waves & Oscillations, Heat, Thermodynamics and Radiation)			

Course Objectives:

Students will able to be familiar with different physical concepts relating to the experiments with complete understanding, realize the applications of these concepts such as mechanics, properties of matter and become confident with the related theories. The aim of this Laboratory is to understand some of the basic phenomenon of mechanics through various experiment. Another prime objective of the course is to enhance the scientific data collection and analysis in Physics Laboratories.

Course Learning Outcomes: Students will be able to

CLO1: Understand different quantities and determination of g

CLO2: (i) Gain the concept Understand measurement techniques of modulus of rigidity.
(ii) Gain the concept of modulus of elasticity.

CLO3: (i)Understand and measure the surface tension of liquids which do wet the contact materials and understand the effect of capillarity.
(ii) Understand and measure the effect of surface tension of a liquid which does not wet the contact material and learn the concept about the effect of cohesive and adhesive forces.
(iii) Learn the concept of surface tension of water, how to form ripples on water surface, and know how to measure the wave length of ripples.

CLO4: Learn the concept of viscosity and capillary flow of water.

CLO5: Understand the techniques to determine specific heat capacity of solid and liquid

CLO6: Measure thermal conductivity of solid

CLO7: Grasp the clear knowledge of temperature coefficient of resistance

CLO8: Understand the laws of transverse vibration of a stretched string by experimentation

CLO9: Determine frequency of a tuning fork by Melde's experiment.

CLO10: Determination of latent heat of fusion of ice and latent heat of condensation of steam.

Mapping of CLOs with PLOs

	PLO-1	PLO-2	PLO-3	PLO-4	PLO-5	PLO-6	PLO-7	PLO-8	PLO-9	PLO10	PLO11	PLO12
CLO1			√	√								
CLO2			√	√								
CLO3			√	√								
CLO4			√	√								
CLO5			√	√								
CLO6			√	√								
CLO7			√	√								
CLO8			√	√								
CLO9			√	√								
CLO10			√	√								

To perform two experiments (one from each group) each of three hours duration.

i) Experiments (3 hours each) 2 x 35 = 70

ii) Laboratory note book 20

iii) Experimental Viva-voce 10

Total marks = 35 Marks for each experiment shall be distributed as follows:

a) Theory 8
b) Data collection and tabulation 12
c) Calculation, graphs and result 10
d) Discussion 5

Total marks = 35

Course Contents (List of Experiments):

Group – A

1. Determination of acceleration due to gravity 'g' by compound pendulum.
2. Determination of acceleration due to gravity 'g' by Kater's pendulum.
3. Determination of Young's modulus and rigidity modulus by Searle's dynamic method.
4. Determination of rigidity modulus of a wire/rod by static method.
5. Determination of rigidity modulus of the material of a wire by dynamic method.
6. Determination of the spring constant and effective mass of a given spiral spring and hence to calculate the rigidity modulus of the material of the spring.
7. Determination of the Young's modulus by the flexure of a beam (bending method).
8. Determination of the moment of inertia of a fly-wheel about its axis of rotation.
9. Determination of the Young's modulus for the material of a wire by Searle's apparatus.
10. Determination of Surface tension of water by capillary tube method.
11. Determination of surface tension of mercury by Quicken's method.

Group – B

12. Determination of the specific heat of solid by method of mixture, with radiation correction.
13. Determination of the specific heat of a liquid by the method of cooling.
14. Determination of the thermal conductivity of a good conductor by Searle's apparatus.
15. Determination of the thermal conductivity of a bad conductor by Lee's method.
16. Determination of mechanical equivalent of heat 'J' with radiation correction.
17. Investigation of the variation of resistance of a copper wire with temperature and determination of its temperature coefficient of resistance.
18. Verify the laws of transverse vibration of a stretched string with a sonometer (n-l, and n – 1/l curves only)
19. Determination of the frequency of a tuning fork by Melde's experiment.
20. Determination of latent heat of fusion of ice with radiation correction.
21. Determination of latent heat of condensation of steam with radiation correction.
22. Determination of density of water at various temperature by specific gravity bottle and study the variation of density with temperature from the graph.

References:

Ahamed, G.U., & Uddin, M.S, (1969). *Practical Physics*. 4th Edition. Hafiz Book Centre.

Din, K., & Matin, M.A. , (1966). *Advanced Practical Physic.* Mallik Brothers.

Khan, A.H., (2019). *B.Sc. Hons. Practical Physics.* Ideal Books.

Worsnop B.l. & Flint, H.T., (1957). *Advanced Practical Physics*. Methuen and Co. Ltd.

Course Code	**213709**	**Marks: 100**	**Credits: 4**	**Class Hours: 56**
Course Title:	**Foundations s of Mathematics**			

Course Objectives:

The emphasis of course is to equip students with the mathematical and critical skills required in solving problems of interest to physicists. The course will also expose students to fundamental computational physics skills enabling them to solve a wide range of physics problems. The skills developed during course will prepare them not only for doing fundamental and applied but also for a wide variety of careers.

Course Learning Outcomes:

Students will be able to

CLO1 Understand the number system and theory of equations which have applications in problems

CLO2 Describe relations between roots and coefficients, Symmetric functions of roots

CLO3 Learn the System of linear equations and basic properties of matrices, different types of matrices viz., Hermitian, skew Hermitian, orthogonal and unitary matrices and their correspondence to physical quantities.

CLO4 Vector Spaces, Linear transformations, Matrix representation of linear transformation, Kernel and image, Eigenvalues and Eigenvectors

CLO5 Explain vector spaces and 2D and 3D geometry.

Mapping of CLOs with PLOs

	PLO-1	PLO-2	PLO-3	PLO-4	PLO-5	PLO-6	PLO-7	PLO-8	PLO-9	PLO10	PLO11	PLO12
CLO1		√									√	√
CLO2		√									√	√
CLO3		√									√	√
CLO4		√									√	√
CLO5		√									√	√

Course Contents:

Topic	Teaching Learning Strategy	CLOs
Relations and functions: Relations; Order relation; Equivalence relations; Functions; Images and inverse images of sets; Injective, surjective, and bijective functions; Inverse functions.	Whiteboard-based lecture Assignment Short presentations by students	CLO1
Real Number System: Field and order properties; Prime numbers; Natural numbers; Integers and rational numbers; Absolute value and its properties; Basic inequalities (including inequalities of means and powers); Inequalities of Cauchy, Chebyshev, Weierstrass.	Whiteboard-based lecture Assignment	CLO2
Complex Number System: Field of Complex numbers; De Moivre's theorem and its applications.	Whiteboard-based lecture Assignment	CLO2
Matrices and Determinants: Notion of matrix; Algebra of matrices; Different types of matrices; Invertible matrices; Determinant function; Properties of determinants; Minors, Cofactors, expansion, and evaluation of determinants. Elementary row and column operations and row-reduced echelon matrices, Invertible matrices, Diagonal, triangular, and Symmetric matrices.	Whiteboard-based lecture Assignment	CLO3
System of Linear Equations: Linear equations; System of linear equations (homogeneous and non-homogeneous) and their solutions using different methods. Gaussian elimination, Application of matrices and determinants for solving systems of linear equations, Applications of systems of equations in real-life problems.	Whiteboard-based lecture Assignment	CLO3

Summation of finite series: Arithmetic and geometric series; Method of difference; Successive differences; Summation of trigonometric series.	Whiteboard-based lecture Assignment	CLO4
Theory of Equations: Relations between roots and coefficients; Symmetric functions of roots; Sum of the powers of roots; Synthetic division; Descartes' rule of signs; Multiplicity of roots; Transformation of equation.	Whiteboard-based lecture Assignment	CLO3
Two-dimensional Geometry: Transformation of coordinates, Pair of straight lines (homogeneous second-degree equations, General second-degree equations representing pair of straight lines, angle between pair of straight lines, Bisectors of angle between pair of straight lines), General equations of second degree (reduction to standard forms, Identifications, Properties and tracing of conics).	Whiteboard-based lecture Assignment	CLO5
Three-dimensional Geometry: Three-dimensional coordinates, Distance, Direction cosines and direction ratios, Planes and straight lines.	Whiteboard-based lecture Assignment	CLO5
Vector Geometry: Vectors in plane and space; Algebra of vectors; Rectangular components; Scalar and Vector products; Coplanar vectors; Scalar triple product and vector triple product; Applications of vectors to geometry (vector equations of straight lines, planes, areas and volumes).	Whiteboard-based lecture Assignment	CLO6 CLO7
Vector Spaces: Euclidean *n*-space, Real vector spaces, Subspaces, Linear combination of vectors, Linear dependence of vectors, Basis and dimension, Linear transformations, Matrix representation of linear transformation, Kernel and image, Eigenvalues and Eigenvectors.	Whiteboard-based lecture Assignment	CLO6 CLO7

References:

Anton, H., & Rorres, C. (2013). *Elementary linear algebra with applications* (11th ed.). Wiley.
Bernet, Cristofer T. (2017). *Geometry*. 6th Edition Schaum's Out line, McGraw Hill .
Bernard, S., and Child, J. M. (2016). *Higher algebra* (6th ed.). Arihant Prakashan.
Hall, H. S., and Knight, S. R. (1992). *Higher algebra* (7th ed.). Macmillan.

Howard A. and Chris R., (2015). *Elementary Linear Algebra with Supplemental Application.* Wiley.
Hummel, J. A. (1994). *Vector geometry*. Prentice Hall.
Khosh M. (2010). *Analytic geometry and vector analysis*. Dhaka: Ideal Library.
Lipschutz, S. (1997). *Schaum's outline of set theory and related topics* (Schaum's Outline Series). McGraw-Hill.
Murrey, R., Spiegel, Seymore L., and Dennis S. (2009). *Vector Analysis*, Schaum's Out line, Springer.
Rahman, M. A. (2015). *Basic algebra*. Dhaka: Nahar Book Depot and Publications.
Rahman, M. A. (2016). *Linear algebra*. Dhaka: Nahar Book Depo t and Publications.
Spiegel, M. R. (1974). *Vector analysis* (Schaum's Outline Series). McGraw-Hill.

Course Code	**213711**	**Marks: 50**	**Credits: 2**	**Class Hours: 28**
Course Title:	**Calculus I**			

Course Description:

Calculus is one of the most fundamental courses in Mathematics, which consists of two parts (Differential and Integral). The course is designed to develop the topics of Differential and Integral calculus. Understanding this course will enable everyone to learn other mathematical courses that require the fundamentals of differentiation and integration.

Objective of the Course:

To develop the basic ideas of functions and their graphs. Learning the basic properties of limit and continuity and analyzing them both mathematically and graphically. To understand the ideas and applications in solving real-life-oriented problems of differentiation and integration.

Course Learning Outcomes:

After completing this course, students will be able to:

CLO1 Identify and graph various types of functions, including polynomial, rational, exponential, logarithmic, trigonometric, and hyperbolic functions, and describe their key properties.

CLO2 Explain the concepts of limits and continuity, and apply relevant theorems to compute limits.

CLO3 Apply techniques of differentiation, including rules and theorems (e.g., Leibniz's rule), to solve problems related to rates of change and approximations in mathematical and real-life contexts.

CLO4 Analyze functions using derivative-based tools, such as the Mean Value Theorem, to determine maximum and minimum values, and concavity, to solve optimization and curve analysis problems.

CLO5 Apply techniques of integration (definite and indefinite) and related theorems to

solve problems involving area, volume, arc length, and surface area.

CLO6 Apply approximation techniques using Taylor polynomials and series to estimate function values and analyze convergence.

CLO7 Evaluate series expansions and perform differentiation and integration of series to solve complex calculus problems.

Mapping of CLOs with PLOs

	PLO-1	PLO-2	PLO-3	PLO-4	PLO-5	PLO-6	PLO-7	PLO-8	PLO-9	PLO10	PLO11	PLO12
CLO1		√									√	
CLO2		√									√	
CLO3		√									√	
CLO4		√									√	
CLO5		√									√	
CLO6		√									√	
CLO7		√									√	

Course Contents:

Topic	Teaching Learning Strategy	CLOs
Functions and Their Graphs: Polynomial and rational functions; logarithmic and exponential functions; trigonometric functions and their inverses; hyperbolic functions and their inverses; combinations of such functions.	Whiteboard-based lecture Assignment	CLO1
Limit and Continuity: Definitions and basic theorems on limit and continuity; Limit at infinity and infinite limits; Computation of limits.	Whiteboard-based lecture Assignment	CLO2
Differentiation: Tangent lines and rates of change; Definition of derivative; One-sided derivatives; Rules of differentiation; Successive differentiation; Leibnitz's theorem; Related rates; Linear approximations and differentials.	Whiteboard-based lecture Assignment Tutorial	CLO3
Applications of Differentiation: Mean value theorem; Maximum and minimum values of functions; Concavity and points of inflection; Optimization problems.	Whiteboard-based lecture Assignment	CLO4
Integration: Antiderivatives and indefinite integrals;	Whiteboard-based	CLO5

Techniques of integration; Definite integration using antiderivatives; Fundamental theorems of calculus; Basic properties of integration; Integration by reduction.	lecture Assignment	
Applications of Integration: Arc length; Plane areas; Surfaces of revolution; Volumes of solids of revolution; Volumes by cylindrical shells; Volumes by cross sections.	Whiteboard-based lecture Assignment	CLO5
Approximation and Series: Taylor polynomials and series; Convergence of series; Taylor's series; Taylor's theorem and remainders; Differentiation and integration of series.	Whiteboard-based lecture Assignment	CLO6 CLO7

Books Recommended:

Anton, H., Bivens, I. C., & Davis, S. (2016). *Calculus: Early transcendental* (11th ed.). Wiley.

Das, B. C., & Mukherjee, B. N. (1938). *Integral calculus*. Kolkata: U. N. Dhur & Sons Pvt. Ltd.

Das, B. C., & Mukherjee, B. N. (1949). *Differential calculus*. Kolkata: U. N. Dhur & Sons Pvt. Ltd.

Matin, M. A., & Chakraboty, B. (1994). *Differential calculus*. Dhaka: Standard Publications.

Mohammad, K., Bhattacharjee, P. K., & Latif, M. A. (1968). *Differential calculus* (1st ed.). Chittagong: S. Tripaty.

Mohammad, K., & Bhattacharjee, P. K. (1987). *Integral calculus* (6th ed.). Chittagong: H. Bhattacharjee.

Stewart, J. (2015). *Calculus: Early transcendentals* (8th ed.). Cengage Learning.

Swokowski, E. W. (1988). *Calculus with analytic geometry* (6th ed.). Brooks/Cole.

Thomas, G. B., & Finney, R. L. (1996). *Calculus and analytic geometry* (9th ed.). Addison-Wesley.

Course Code	Course Title	Marks	Credits	Class Hours
212807	**Chemistry-I**	**100**	**4**	**56**

Course Objective:

The course is designed for the student to impart and improve fundamental knowledge and aspects related to understand the fundamental physical, Organic and Inorganic chemistry topics. To develop and extend student's knowledge on state of matter, structure of atom, radiochemistry, chemical bonding, Liquid and Solution, chemical equilibrium, pH, aliphatic and aromatic hydrocarbons.

Course Learning Outcomes (CLOs)

After completing this course, students will be able to:

CLO1 Explain the historical background, fundamental concept, and scope and aim of concepts of physical, Organic and inorganic chemistry.

CLO2 Clarify the state of matter, structure of atom, radiochemistry

CLO3 Explain about the chemical bonding, Liquid and Solution

CLO4 explain the chemical equilibrium, pH, aliphatic and aromatic hydrocarbons

CLO5 Explain the chemical equilibrium, pH, aliphatic and aromatic hydrocarbons, bonding, synthesis, nomenclature and properties.

Mapping of CLOs with PLOs

	PLO-1	PLO-2	PLO-3	PLO-4	PLO-5	PLO-6	PLO-7	PLO-8	PLO-9	PLO10	PLO11	PLO12
CLO1	√	√										
CLO2	√	√										
CLO3	√	√										
CLO4	√	√										
CLO5	√	√										

Course Contents:

	Topics	**Teaching Learning Strategy**	**CLOs**
1	**State of Matter in Chemistry:** Nature and classification of matter. Classical states of solid, liquid, gas and Liquid crystals, state of aggregation of matter. Changes of state of matter. Intramolecular and intermolecular force and their role for the determination of state of matter. Modern state as plasma. Avogadro's number with application, Concept of mole: calculation and importance. Stoichiometry.	Lecture, Group Discussion, Assignment	1, 2, 3
2	**Structure of atom:** Fundamental particles of atoms, Nucleus and discovery of nucleus, atomic model of Rutherford, Bohr atom model: postulates, limitations and success, Spectrum of atomic hydrogen, Dual nature of	Lecture, Group Discussion, Assignment	3, 4, 5

	electron, Heisenberg uncertainty principle, Quantum numbers, atomic orbitals, Aufbau principle, Pauli exclusion principle, Hund's rule of maximum multiplicity, electronic configuration of atoms.		
3	**Radiochemistry:** Atomic number and mass number, atomic mass unit, atomic nucleus, Isotopes, Nuclear stability, natural and artificial radioactivity, half life and average life of radioelements, radioactive decay, nuclear reactions.	Lecture, Group Discussion, Assignment	3, 4, 5
4	**Periodic table:** Periodic law, Periodic table, Periodic properties of the elements such as ionization energies, Electron affinity, Electro negativity, Atomic/ionic radius along a period and down a group, Diagonal relationship.	Lecture, Group Discussion, Assignment	2, 3, 4
5	**Chemical bond:** Definition and causes chemical bonds, octet rule and duplet rule, types of chemical bonds: ionic, covalent bond, polar and non-polar bond, Ionic character of covalent compound and covalent character of ionic compounds VSEPR theory, Valence bond theory, Hybridization, σ and π-bonding in compounds, Molecular orbital theory, coordination, metallic, hydrogen bonds.	Lecture, Group Discussion, Assignment	2, 3, 4, 5
6	**Chemical Reactions:** Physical change and Chemical change, Methods of bringing about chemical reaction, Redox reactions, oxidizing and reducing agents, oxidation state and oxidation number, redox half reactions, rules for balancing redox reactions, acid base reactions.	Lecture, Group Discussion, Assignment	2, 3, 4
7	**Gaseous State:** Vapor and gas, measurable properties of gases, determination of gas pressure, the gas laws: derivation, molecular explanation and practical applications, effusion, diffusion, osmosis, ideal and real gases, ideal gas equation with applications, kinetic theory of gas, deviation from ideal behavior, Van der Walls equation, critical constants. Dalton's law of partial pressure.	Lecture, Group Discussion, Assignment	1, 3

8	**Liquids and Solutions:** Liquids and their characteristics, evaporation, measurement vapor pressure, dependence of vapor pressure on temperature variation, boiling, distillation, Types of solution, solubility and solubility curve, units of concentration for solution, raoult's law, ideal and non-ideal solutions. Henry's law. a brief treatment of colligative properties.	Lecture, Group Discussion, Assignment	4, 5
9	**Chemical Equilibrium:** Equilibrium in chemical reactions. Stoichiometry, law of mass action, concentration versus time curves, equilibrium law and equilibrium constant: Kp, Kc, and Kx, effects of temperature, pressure and concentration changes of equilibrium, principle and applications of Le Chatelier and Braun, reaction quotient, extent of reaction, thermodynamic equilibrium constant.	Lecture, Group Discussion, Assignment	1
10	**pH and Buffer:** Ostwald dilution law, solubility product with principle and applications, common ion effect, ionic product of water (K_W), pH with applications, buffer solution, buffer mechanism, Henderson-Hasselbach equation. Acid-base titrations, Acid-base indicators. Theories of acids and bases, conjugate acids and bases, neutralization reactions, acid-base strength, Salts: Classification and their applications, neutralization reactions.	Lecture, Group Discussion, Assignment	1, 2
11	**Aliphatic hydrocarbons:** Organic compounds, homologous series, the causes of diversity of organic compounds, Hydrocarbons: classifications and IUPAC nomenclature. Formation of carbon cations, carbanion, free radicals and their stabilities. Sources, properties, general methods of preparation of following hydrocarbons: Alkane, Alkene and Alkynes. Classification of organic compounds based on their functional groups.	Lecture, Group Discussion, Assignment	5
12	**Aromatic hydrocarbons:** Definition and classification, nucleus and side chain, Structure, preparation and uses of benzene. Aromaticity, Electrophilic aromatic substitution reactions with reference to nitration, halogenation,	Lecture, Group Discussion, Assignment	5

	sulphonation and alkylation. Orientation of substituents in benzene ring, activating and deactivating group.		

References:

Bahl, B. S., & Bahl, A. (2024). *Text- Book of Organic Chemistry*. S. Chand & Company Ltd Ram Nagar, New Delhi

English, J., & H.G Cassidy and Baird, R. I. (1949). *Principles of Organic Chemistry: An Introductory Text in Organic Chemistry*. McGraw Hill.

Finar, I. L. (1973). *Organic Chemistry, Vol. I*. Pearson Education India.

Griffin, R. W. (2024). *Modern Organic Chemistry*. McGraw Hill.

Morrison, R.T. & Boyd, R. N. (1992). *Organic Chemistry*. Benjamin-Cummings Pub Co

Muny, J. Mc. (2010). *Organic Chemistr.* Thomson Broks-Coole.

Robert, J. D., & Caserio, M. C. (1977). *Basic Principles of Organic Chemistry*. W. A. Benjamin, Inc., Menlo Park, CA.

Solomons, T. W. G. (2008). *Fundamental of Organic Chemistry*. Wiley.

Course Code	Course Title	Marks	Credits	Class Hours
212808	**Chemistry-I Practical**	**50**	**2**	**28**

Course Objectives:

To gain a fundamental knowledge on laboratory safety policy, some inorganic salts and radical analysis, preparation of solution with standardization, titration and precipitation.

Course Learning Outcomes (CLOs)

After completing this course, students will be able to:

CLO1 Discuss about the general laboratory safety policy, rules and regulations and chemical management process.

CLO2 Explain the applications of apparatus and reagents in various experiments, different laboratory glassware and chemicals.

CLO3 Prepare stander solution and pH- neutralization curves

CLO4 Qualitative analysis of mixtures of inorganic salts consisting of up to five different radicals

CLO5 Explain to determination of Fe^{2+}, copper (II), nickel as Ni (HDMG) 2 complex 7 using different methods.

Mapping of CLOs with PLOs

	PLO-1	PLO-2	PLO-3	PLO-4	PLO-5	PLO-6	PLO-7	PLO-8	PLO-9	PLO10	PLO11	PLO12
CLO1			√	√	√						√	
CLO2			√	√	√						√	
CLO3			√	√	√						√	
CLO4			√	√	√						√	
CLO5			√	√	√						√	

Course Contents

	Topics	Teaching Learning Strategy	CLOs
1	**Safety:** Introduction of chemical, equipment's and safety in the laboratory.	Lecture, Group Discussion, Experimental, Assignment	1-5
2	**Preparation:** Preparation of $FeSO_4.7H_2O$ Mohr's salt and potash alum.	Lecture, Group Discussion, Experimental, Assignment	1-5
3	**Separation and identification**: Separation and identification of four radicals from a mixture of anions and cations The cations are Pb^{2+}, Cu^{2+}, Cd^{2+} , Al^{3+} , Fe^{2+} , Fe^{3+}, Co^{2+}, Ni^{2+}, Zn^{2+} Ca^{2+}, Ba^{2+}, Na^{+}, K^{+} , and NH_4^{+} , the anions are NO_3^{-}, CO_3^{2-}, S^{2-}, SO_4^{2-}, Cl^{-} , Br^{-} and I^{-}.	Lecture, Group Discussion, Experimental, Assignment	1-5
4	**Standardization:** Standardization of NaOH solution using standard oxalic acid solution.	Lecture, Group Discussion, Experimental, Assignment	1-5

5	**Determination:** Determination of Fe^{2+} using standard permanganate solution.	Lecture, Group Discussion, Experimental, Assignment	1-5
6	**Iodometric method:** Determination of copper (II) using standard Na_2SO_3 solution.	Lecture, Group Discussion, Experimental, Assignment	1-5
7	**Gravimetric Method:** Gravimetric determination of nickel as Ni $(HDMG)_2$ complex.	Lecture, Group Discussion, Experimental, Assignment	1-5
8	**pH-Neutralization Curves:** Determination of the pH-neutralization curves of a strong acid by a strong base.	Lecture, Group Discussion, Experimental, Assignment	1-5

References:

Alexeyev, V. (2024). *Qualitative Analysis*. Mir Publishers.

Braun, D. R. (2024). *Introduction to Chemical Analysis*. McGraw Hill International,

Christian, G. D. (2023). *Analytical chemistry*. John Wiley & Sons.

Fifield, F. W., & Kealey, D. (2000). *Principles & practice of analytical chemistry*. Wiley.

Skoog, D. A., & West, M. D. (2022). *Fundamental of analytical chemistry*. Publisher, Cengage Learning.

Vogel, A. I., (2017). *A Text-Book of Macro and Semimicro Qualitative Inorganic Analysis*. Longmans, Green and Co. Ltd.

Welcher, F. J., & Hahn, R. B. (2024). *Semimicro Qualitative Analysis*. D. Van Nostrand Co. Inc.

Wulfsberg, G., (1987). *Principles of Descriptive Inorganic Chemistry*. University Science Books, Mill Valley.

Course Code	213607	Marks:100	Credits: 4	Class Hours: 56
Course Title:	Fundamentals of Statistics			

Course Objectives:

To be able to understand the nature, characteristics, scope, application and abuse of statistics. To make familiar with data, nature of data, how to process and condense the data, sources of data and graphical presentation of data, and to apply appropriate statistical tools and techniques to analyze the data. To acquaint students with necessary skills for solving probability related problems using appropriate laws. To provide knowledge on time series and statistical indices.

Course Learning Outcomes:

After going through the course, students will be able to learn the following outcomes

CLO1 Understand the nature, characteristics, scope, application and abuse of statistics. Knowledge on sources of data and how to process, condense and presentation of data. Understand different characteristics of statistical data such as measures of location, dispersion, moments, skewness, kurtosis and their properties.

CLO2 Understand the relationship between variables such as simple correlation, rank correlation, correlation ratio, simple regression analysis. standard error of estimators of regression coefficients & their properties and fitting of regression lines.

CLO3 Comprehend different approaches of defining probability and useful laws of probability to solve problems. Also learn some commonly used probability distributions.

CLO4 Gain knowledge about meaning and application of statistical indices.

CLO5 Identify the pattern and trends and isolate the influencing factors of the time series data for future planning and control.

CLO6 Understand the concept of numerical methods, including interpolation and use of numerical methods in application to real problems.

Mapping of CLOs with PLOs

	PLO-1	PLO-2	PLO-3	PLO-4	PLO-5	PLO-6	PLO-7	PLO-8	PLO-9	PLO10	PLO11	PLO12
CLO1			√	√	√						√	
CLO2			√	√	√						√	
CLO3			√	√	√						√	
CLO4			√	√	√						√	
CLO5			√	√	√						√	
CLO6			√	√	√						√	
CLO7			√	√	√						√	

Course Contents:

Topic	Teaching Learning Strategy	CLOs
Descriptive Statistics: Statistics–Its nature and some important uses, Qualitative and quantitative data, Classification, Tabulation and frequency distribution, Graphical representation of data, Measures of location, Measures of Dispersion, Skewness and Kurtosis, Mathematical relationship among different measures of location, dispersion, Skewness and kurtosis.	Lecture, Assignment	CLO1
Bivariate Data: Correlation coefficient, Correlation analysis, The purpose and uses of regression analysis, Simple regression and methods of least squares and estimation of parameters, Correlation ratio, Rank correlation, Partial and multiple correlation.	Lecture, Assignment	CLO2
Elementary Probability: Meaning of Probability, Classical and empirical definitions of Probability, Axiomatic approach of defining probability, Event, Sample space and simple problems on probability, Addition rule, Conditional probability, Multiplication rule and Bayes theorems, The concept of a random variables, Probability function and probability density function, Joint probability function. Marginal and conditional distributions, Statistical independence, Expected value and related theorems,	Lecture, Group Discussion, Assignment	CLO3

Moment generating function, Common probability distributions, Binomial, Poisson and Normal.		
Index Number: Concept of an index number and problems in the construction of index number, Types of indices (Price, Quantity, Value and cost of living indices) and their uses, Tests for index numbers.	Lecture, Group Discussion, Workshop and Assignment	CLO4
Time Series analysis: Elements of time-series analysis, Measurement of trend by moving average, By least square method, Trend curve, Determination of seasonal indices, Cyclical movements.	Lecture, Group Discussion, and Assignment	CLO5
Numerical Mathematics: Differences of a polynomial, Finite difference operator, Difference table, Newton's formula and starling's central difference formula, Inverse interpolation, Numerical integration.	Lecture, Group Discussion, Workshop and Assignment	CLO6

References:

Gupta, S.C., & Kapoor, V.K. (1994). Fundamentals of *Applied Statistics*. Sultan Chand & Sons.

Hoel, P.G. *Introduction to Statistics, 4th Edition*. Wiley and Sons

Islam, M.N. (2015). *An introduction to Statistics and Probability (4th ed.).* Mullick & Brothers.

Jalil, M. A. and Ferdous R. (1999). *Basic statistics: Methods and Applications*, Robi Mostafa, M.G. (1989). *Method of Statistics (4th ed.).*Karim press and Publications.

Shil, R.N., & Debnuth, S.C. (2016). *An introduction to the theory of Statistics*. Star Publications.

Weiss, N. A., & C. A. Weiss (2012). *Introductory statistics*. Pearson Education.

Course Code	213610	Marks:50	Credits: 2	Class Hours: 28
Course Title:	Lab-1 : Fundamentals of Statistics			

Data condensation and tabulation. Formation of frequency distribution from both qualitative and quantitative data. Construction of bivariate table. Graphical representation of data. Measures of location and dispersion, Calculation of moments, Measures of skewness and kurtosis. Simple correlation coefficient and fitting of regression lines. Computation of rank correlation coefficient. Fitting of Binomial, Normal and Poisson's distributions, Finding trend values and seasonal variation from time series data by different methods, Calculation of Index numbers and test of index number, Use of Newton's forward and backward formula, Solution of numerical integration.

Course Code:	219901	Marks: 100	Credits: 4
Course Title:	History of Bangladesh: Language, Culture, and Identity		Hours: 60

Course Objectives: This course is designed to help undergraduate students from diverse academic backgrounds develop a comprehensive and nuanced understanding of the historical development of Bangladesh, with a particular focus on its language, culture, and identity. By critically examining a variety of historical events, socio-political movements, and cultural shifts from ancient times to the contemporary period, students will have the opportunity to trace the evolution of the Bengali nation. The course aims to foster informed citizenship through an exploration of the Liberation War, identity formation processes, cultural heritage, the lives and contributions of key political figures, and the roles of ethnic minorities. Ultimately, the objective is to equip learners with the intellectual tools necessary to contextualize current national debates and to recognize the role of youth and globalization in shaping the future of Bangladesh.

Course Learning Outcomes (CLOs): At the end of the course, learners will be able to
CLO 1: Recall key historical events, dates, movements, personalities, and cultural developments that contributed to the emergence of Bangladesh from the pre-colonial period to the present day.
CLO 2: Demonstrate an understanding of the socio-political, linguistic, and cultural influences that shaped Bengali identity and nationalism, emphasizing the Language Movement, Liberation War, and subsequent democratic uprisings.
CLO 3: Apply their historical and cultural understanding to interpret contemporary national issues and debates relating to identity, political reform, and cultural transformation in Bangladesh.
CLO 4: Analyze how historical events, political ideologies, and social changes are interconnected in shaping the current socio-political landscape of Bangladesh, with a particular emphasis on the roles of movements, minorities, and the youth.

Course Contents:

Course contents	Teaching Learning Strategy	CLOs
1. Pre-colonial Era • Life and Culture of the People in Ancient Bengal • Bengal Under Muslim Rulers: Society, Culture, and Religion • Role of the Sufis in Preaching Islam and Impacts of • Sufism on the Bengali Society • Bengali Society and Culture in the Writings of Foreigners	Interactive lectures	CLO 1-2
2. Colonial Era (18th and 19th Century) • The Battle of Plassey (1757) and the Beginning of British Colonialism • Bengal Renaissance • Reforms in Hindu Society • New Forms in Bengali Literature and Culture • The Muslim Response to Western Education	Interactive lectures, reading, and assignments	CLO 1-2
3. Colonial Era (First Half of the 20th Century) • Partition of Bengal (1905) • Hindu-Muslim Disagreements • Formation of the All India Muslim League (AIML) • Muslim Shahitya Samaj • Buddhir Mukti Andolan: The Urge for Rational Thinking in Bengali Muslim Society • Growth of Religion-based Identity • Politics of Hindutva and the Two-Nation Theory • Spread of Communalism in Society • The Partition of India and Bengal	Interactive lectures and group discussions	CLO 2-3
4. Post-Partition Era (1947-1971) • Language Movement • Political, Economic, and Cultural Aspects • Growth of Vernacular Nationalism • Cultural Activism	Interactive lectures and group discussion	CLO 2-3
5. Changing Bengali Identity • The Evolution of Bengali Identity in the Context of Language, Culture, and Religion • From Ancient Times to the Present • The Role of the Bengali Language in Shaping Identity • The Language Movement of 1952 and Its Long-term Impact on National • Consciousness • The Influence of Religion on Bengali Identity • The Impact of Socio-political Movements on Identity	Interactive lectures and thematic assignments	CLO 2-3

Formation • The Liberation War of 1971 • Contemporary Debates on Bengali Identity • The Role of Youth in Redefining Identity • The Influence of Globalization		
6. Liberation Movement of 1971 and Mass Uprising till 2024 • The Political and Economic Exploitation of East Pakistan by West Pakistan • Six (6) Points Movement, Uprising of 1969 • The 1970 General Elections • The Non-cooperation Movement and the Declaration of Independence • The Role of AK Fazlul Haque, Huseyn Shaheed Suhrawardy, Maulana Bhashani, Sheikh Mujibur Rahman, and Ziaur Rahman • The Liberation War of 1971 • Genocide and Resistance • The Role of the Mukti Bahini • Post-independence Challenges • Nation-building • Political Instability • Mass Uprisings and Democratic Movements • The Anti-autocracy Movement of the 1980s • The 1990s Movement for Democracy and Afterwards • The 2024 Movements for Political and Social Reforms	Interactive lectures, group discussions, and thematic assignments	CLO 1 CLO 3-4
7. History of Other Ethnic Groups • The Indigenous Communities of Bangladesh • Historical Presence • Cultural Practices • Land Rights and Cultural Assimilation • Contributions and Sacrifices • Political Representation of Ethnic Minorities	Interactive lectures and group discussions	CLO 4
8. Cultural Heritage and Modern Transformations • The Evolution of Bengali Culture • From Ancient Traditions to Modern Expressions • The Role of Literature, Music, and Art in Shaping Bengali Identity • Contributions of Rabindranath Tagore, Kazi Nazrul Islam, and Other Cultural Icons • The Impact of Globalization on Bengali Culture • The Revival of Traditional Arts • The Role of Youth in Cultural Innovation Urbanization • Environmental Changes • The Commodification of Culture	Interactive lectures, documentary screening, and thematic assignments	CLO 1-4

List of Books:

Ahmed, M. (1979). *Bangladesh: The constitutional quest for autonomy*. University Press Limited.

Ahmed, R. (1981). *The Bengal Muslims 1871–1906: A quest for identity*. Oxford University Press.

Alavi, H. (1972). The state in post-colonial societies: Pakistan and Bangladesh. In K. Gough & H. P. Sharma (Eds.), *Imperialism and revolution in South Asia* (pp. 145–178). Monthly Review Press.

Bernier, F. (2023). *Travels in the Mogul Empire: A.D. 1656–1668*. Atlantic Publishers.

Bleie, T. (2005). *Tribal peoples, nationalism, and the human rights challenge: The Adivasis of Bangladesh*. University Press Limited.

Bose, S. (2011). *Dead reckoning: Memories of the 1971 Bangladesh war*. Hurst & Company.

Eaton, R. M. (1996). *The rise of Islam and the Bengal frontier, 1204–1760*. University of California Press.

Gilmour, D. (2019). *The British in India: Three centuries of ambition and experience*. Penguin.

Habib, L. (1982). *The Cambridge economic history of India*. Cambridge University Press.

Halim, S., Amanullah, A. S. M., & Nasir, R. I. (Eds.). (2024). *Society and sociology in Bangladesh: A South Asian perspective*. University Press Limited.

Hashmi, T. (2021). *Fifty years of Bangladesh, 1971–2021: Crises of culture, development, governance and identity*. Palgrave Macmillan.

Huq, M. E. (1975). *History of Sufism in Bengal*. Bangladesh Asiatic Society.

Husain, I. (2014). *Karl Marx on India*. Tulika Books.

Jahan, R. (1972). *Pakistan: Failure in national integration*. Columbia University Press.

Maniruzzaman, T. (1988). *The Bangladesh revolution and its aftermath* (2nd ed.). University Press Limited. (Original work published 1980)

Mascarenhas, A. (1986). *Bangladesh: A legacy of blood*. Hodder and Stoughton.

Rashid, H. O. (2015). *The foreshadowing of Bangladesh: Bengal Muslim League and Muslim politics* (3rd ed.). University Press Limited.

Roy, A. (1984). *The Islamic syncretistic tradition in Bengal*. Princeton University Press.

Sen, A. (2006). *Identity and violence: The illusion of destiny*. W. W. Norton & Company.

Umar, B. (2022). *The emergency of Bangladesh: A history of East Pakistan*. Bangla Gobeshona.

Course Code	**219903**	**Marks: 75**	**Credits: 3**
Course Title:	**Information and Communication Technology**		

Course Objectives:

The main objective of the course is to develop students' understanding and skills in using, managing, and applying technology to solve problems and enhance various aspects of life and

work. Key areas of focus include understanding ICT systems, software, hardware, networks, and their applications.

Course Learning Outcomes: The student will be able to:

CLO 1	Explain foundational ICT concepts, including the information processing cycle.
CLO 2	Operate standard computer hardware and software systems effectively.
CLO 3	Use office productivity tools (Word, Excel, PowerPoint) for academic and professional tasks.
CLO 4	Apply safe internet practices and use internet tools for communication and information retrieval.
CLO 5	Understand and explain the concepts of Data Analytics, Artificial Intelligence (AI), and Machine Learning (ML)
CLO 6	Understand the concept of free-lancing

Course Contents:

Unit	Specific Objectives	Content	Teaching and Learning Approach
Unit 1: Introduction to Information and Communications Technology (ICT) and Computer System	The student will be able to: • Explain the concept of ICT and its related terminologies. • Describe the information processing cycle. • Analyze the impact of ICT on educational, social and economic development. • Identify career opportunities in ICT education. • Differentiate among the classes of computers and the usages of them. • Identify the vital components of the Systems Unit.	Definition of ICT, basic concepts and terminologies. Data, Information Application of ICT, Advantages and Disadvantages of ICT, Impact of ICT Career opportunities in ICT education Information Processing Cycle. Classification of Computers The Vital Components of the Systems Unit.	Classroom Lectures
Unit 2:	The student will be able to:	Categories of Computer	Classroom

Computer Hardware and Software	• Describe the categories of Computer Hardware. • Describe the commonly used Input and Output devices • Identify the main processing devices, storage devices and media. Identify the main communication devices. • Identify types of Software packages. • Distinguish between an Operating System and Application software. • Identify different types, examples and uses of Operating Systems and Application software • Distinguish between Open Source and Proprietary Software.	Hardware: Input devices, Processing devices, Output devices, Storage devices Communication devices, Main Processing Devices: The Processor, Control Unit and Arithmetic and Logic Unit Software Packages, Operating Systems, Types and uses of Operating Systems, Types and uses Application Software: Educational software, Games software, Graphics software Browsers: Internet explorer, Google chrome, Mozilla Firefox, Opera, Internet explorer, Mozilla Firefox; Proprietary and Open-Source Software	lectures and Lab
Unit 3: Introduction to Word Processing Application	• The student will be able to: • Identify Word Processing packages. • Create and save a document using the Word Processor. • Format a Word document using formatting tools. Demonstrate the ability to perform collaborative editing. • Insert tables in a Word Processing document. • Insert symbols and pictures in Word Processing documents. • Use layout techniques in document creation. • Inserting headers and footers. • Print documents using the various print options.	Word Processing Packages and Their Uses. Introduction of Microsoft Word Software. File creation and saving: New, Save, Open, Save As. Editing: Copy/Cut, Paste, Undo, Redo, Find, Replace, Clipboard, Sub-Headings And Paragraphs, Text Correction, Wrapping Text, Text Orientation **Formatting:** Font: Style, Size, Color, Bold, Underline, Italic, Superscript, Subscript, Shadow, Strikeout. Paragraph: Alignment, Bullet & Numbering, Indent, Line Spacing, Table Border **Insert:** Table, Picture, Shape,	Classroom Lectures, Lab, and Hands-on Practice

		Smart Art, Chart, Header, Footer, Page Number, WordArt, Drop Cap, Symbols, Equations, Hyperlink, Comment. **Page Design:** watermark, page border **Layout:** Page setup, print	
Unit 4: Spreadsheet Application	• The student will be able to: • Identify Spreadsheet Packages. • Explain the importance of the Spreadsheet application in data management. • Explain related concepts and terminologies in the Spreadsheet. • Identify features in the Spreadsheet application window. • Create and save a Workbook. • Construct and insert simple formulae and functions. • Format the worksheet using formatting tools. • Printing a worksheet.	Spreadsheet Packages, e.g. Excel. Importance of Spreadsheet application in Data Management, Related Concepts and Terminologies (e.g., cell(s), rows, columns, worksheet, workbook) Excel Tool Bars: Formatting Bar, Standard Bar, Formula Bar. Types of Data and Their Uses (e.g., number, date, text, currency), Creating and Saving a Workbook, Formula and Functions, Sorting, filtering, validation, creating pivot table. Formatting Worksheet Using Formatting Tools. Draw Graph/chart, Printing Worksheet.	Classroom Lectures, Lab and Hands-on Practice
Unit 5: Presentation Application	• The student will be able to: • State the importance of the Presentation application. • Identify the commonly used features of a Presentation application while prepare a presentation. • Create and save presentations using a template, Add new slide(s). Edit text, Format	Presentation Applications Packages, Devices used for Presenting, Importance of Presentation Application, Principles for Designing Presentations, Terminologies in Presentation Application (eg;, Slide transitions, Slide show, etc.), Commonly Used Features of	Classroom Lectures, Lab and Hands-on Practice

	text, Insert objects, images and pictures, Run slide show, Apply transition, animation effects to slides • Select the print option for printing. • Prepare a presentation on a selected topic and present it.	Presentation Application Window: Toolbars, Different presentation view modes Prepare a Presentation: Slide creating, editing, theme, background, inserting elements (image, audio, video hyperlink, table, smart art, graph, etc.) and formatting slides, Slide Layout, slide Show, slide transition, animation Effects. Selection of Print Option: Entire presentation, Specific slides, Handouts, Notes pages, outline view of slides.	
Unit 6: Privacy and Security	The student will be able to: • Understand the basics of digital security • Use some security tools. • Understanding digital ethics.	Introduction to Information Security, cybercrime, DoS and DDoS Attack, Key Management, Digital Signature and Certifications, privacy, Data Security, Vulnerability, Threat and Risk, Malware, Social Engineering, Hacking, Plagiarism, Fishing, Software Piracy, Worms and Viruses, Spam, Adware, Malware, Spyware, Antivirus Software Ethics in the digital world	Classroom Lectures, Lab and Hands-on Practice
Unit 7: Using The Internet to Communicate and Accessing Information	• The student will be able to: • Explain basic concepts, requirements, and terminologies of the Internet • Apply the rules and regulations in the use of the internet. • Using email • Use the internet social network to communicate. • Use Uniform Resource Locators (URLs) to access Information. Use search engines to access	Internet, Intranet, Extranet, IP Address, Masking, MAC Address, Internet Services, OSI Reference Model, TCP/IP protocol stack, IPv4, IPv6, subnet Masking, MAC Address, Internet Services, Network Configuration and Troubleshooting, Wi-Fi, Broadband, Email Usage. Rules and Regulations in the Use of the Internet: Spam-Unsolicited Emails, People's Privacy, Intellectual Property	Classroom Lectures, Lab and Hands-on Practice

	information • Upload files to virtual drives and work on it.	Rights, etc. E-mail: Creating an Email Account, Sending, Accessing Email Messages, Attaching Documents to Email Messages, Using the Internet: Social Networks to Communicate, Uniform Resource Locators (URLs) to Access Information, Using Search Engines, Downloading Information from the Internet. Transferring Information from the Internet to a Different Application Intellectual Property Rights, ICT Policy, Software Piracy etc. Using Cloud Space: Google Drive, Google Workspace, OneDrive, Dropbox, etc.	
Unit 8: Emerging Technologies: Data analytics, Artificial intelligence, Machine learning	• Define Data Analytics, Artificial Intelligence (AI), and Machine Learning (ML). • Understand how data is collected, processed, and used for decision-making. • Recognize the role of AI and ML in everyday life. • Explain basic differences between AI and ML. • Discuss benefits and challenges of these technologies. • Recognize current trends and career opportunities in these fields.	**Introduction to Data Analytics:** Define Data, Types of Data, Define Data Analytics, Basic Steps in Data Analytics, Applications of Data Analytics, Common Tools of Data Analytics (Excel, Tableau, Power BI, Python, SQL, etc.). **Artificial Intelligence (AI):** Define AI, Objectives of AI, AI capabilities, AI applications (NLP, Image Recognition, Predictive Analytics, Recommender Systems, Robotics, etc.); **AI Tools**: OpenAI GPT, TensorFlow, PyTorch, Keras, Scikit-learn, etc.), AI in daily life (Voice assistants, facial recognition, GPS route suggestions, chatbots, etc.).	Classroom Lectures, Lab and Hands-on Practice

		Machine Learning (ML): Define Machine Learning (ML), How It Works, Classification of ML, applications of ML, and common ML platforms **(**Teachable Machine, TensorFlow, PyTorch, Microsoft Azure Machine Learning, etc.**).** Relationship Between Data Analytics, AI, and ML Benefits & Challenges of Data Analytics, AI, and ML. Future Trends & Career Paths.	
Unit 9: Introduction to Freelancing	The student will be able to: • Define and explain the fundamental concept and significance of freelancing in the global context. • Identify and categorize various online marketplaces and professional fields. • Develop professional profiles, portfolios, and proposals suitable for freelancing platforms. • Demonstrate the use of online payment systems and financial transaction methods. • Analyze and evaluate strategies for career development and effectively apply techniques to overcome	Introduction and Basic Concepts, Fields of Freelancing, Essential Skills and Preparation, Introduction to Freelancing Marketplaces, Creating Profile and Portfolio, Earning and Transaction Process, Building a Freelancing Career, Challenges and Solutions.	Classroom Lectures, Practical Demos, Assignments, Discussions

	freelancing challenges.		

List of Books:

Norton, P. (2008). *Introduction to computers* (9th ed.). The McGraw-Hill.

Comer, D. E. (2018). *The internet book: Everything you need to know about computer networking and how the internet works* (5th ed.). Chapman and Hall/CRC Press.
Lambert, J., & Frye, C. (2015). *Microsoft Office 2016 step by step*. Microsoft Press.
Hassan, N. A., & Hijazi, R. (2017). *Digital privacy and security using Windows: A practical guide*. Apress.
Maheshwari, A. (2024). *Data analytics made accessible*. Jay Cobb.
Russell, S. J., & Norvig, P. (2020). *Artificial intelligence: A modern approach* (4th ed.). Pearson.
Alpaydin, E. (2020). *Introduction to machine learning* (4th ed.). MIT Press.

Course Code	**219904**	**Marks: 25**	**Credit: 1**
Course Title:	**Lab: Information and Communication Technology**		

Course Objectives:

The main objective of the course is to teach the students' understanding and skills in using, and applying technology to solve problems and enhance various aspects of life and work. It includes assembling hardware, installing software, preparing PPT slides, and producing Word and Excel documents.

Course Learning Outcomes:

The student will be able to:

CLO 1	Learn assembling hardware
CLO 2	Prepare, edit and print Word documents and excel
CLO 3	Prepare Power Point presentation.
CLO 4	Access information from Internet, e-mail
CLO 5	Installation of anti-virus software
CLO 6	Data collection, Prediction, Pattern recognition, classification
CLO 7	Exploring markets, Proposal writing, Simulation strategies

List of Experiments

CLO Addressed	Unit	Experiments	Teaching Learning Approach
CLO1	1-4	• Assemble different hardware • Install different software • Operate the computer - Drive, folder and file management • Maintenance	Lab and Hands-on Practice
CLO 2	3	**Word** • Prepare a Word document on a specific topic (e.g, routine, question paper, CV, reports, applications) • Formatting the document (Alignment, table, border, watermark, etc.), e.g., newspaper article, academic report, or documentation used in daily life, book, poster • Print documents with different paper and printers	Lab and Hands-on Practice
CLO 2	4	• **Excel** • Prepare a grade sheet • Prepare a family expenditure • Prepare a business expenditure report • Prepare payroll management, with a report • Create graphs on the given data • Print Excel files	Lab and Hands-on Practice
CLO 3	5	**Power point** • Prepare an academic presentation on a specific topic. • Formatting the slides & using different tools. • Apply animation and transition • Print PPT files in different modes: Hand note, Slides shorter, outline	Lab and Hands-on Practice
CLO4	6	• Install antivirus software, e.g., Norton Antivirus, McAfee, Kaspersky, Avast.	Lab and Hands-on Practice
CLO5	7	• Use of email • Access information from the internet, use a search engine. • Use of virtual drive for collaboration • Google Meet	Lab and Hands-on Practice
CLO6	8	• Data Collection and Visualization (Excel) • Simple Prediction (Excel) • Pattern recognition (Teachable Machine)	Lab and Hands-on Practice

		• Classification (Teachable Machine)	
CLO7	9	**Free lancing** • Marketplace exploration and portfolio creation • Proposal writing • Payment simulation	Lab, Hands-on, Group Work, Discussion

Instructions for the Lab:

1. Lab sessions should be hands-on, with students engaging in tasks like assembling hardware, installing software, and using office tools.
2. Software like Microsoft Word, Excel, PowerPoint, and Teachable Machine will be used for document creation, data analysis, and basic machine learning.
3. Labs also include antivirus installation, email use, and online collaboration via Google Meet and Drive, emphasizing teamwork and problem-solving.

Instructions for the Lab exam:

1. The lab exam is individual. Students must answer two out of seven questions.
2. Marks distribution: Experiment (20 marks, 10 for each) for practical tasks, Viva-Voce (5 marks) for oral assessment. Total: 25 marks.

List of Books:

Comer, D. E. (2018). *The internet book: Everything you need to know about computer networking and how the internet works* (5th ed.). Chapman and Hall/CRC Press.

Lambert, J., & Frye, C. (2015). *Microsoft office 2016 step by step*. Microsoft Press